Scheduler Planning

My role:

- Microcontroller used: STC89C52 Microcontroller , STC89C52 is a 8 bit microcontroller. It is based on 8051 architecture. It contains CPU, memory, registers, GPIO ports, timers and interrupts etc. It is used to control different electronic devices.

1.Overview:

The scheduler is mainly responsible for the management of multiple programs as processes and deciding which process gets the CPU. It works with the process management and CPU modules of the educational microcontroller simulator.

2.Objective:

- Manages multiple programs
- Schedules next program
- Demonstrates FCFS,roundrobin,priority schgeduling
- Manages states of execution
- Supports content switching

3.Main algorithms

- FCFS (First Come First Serve): Process selected according to order of arrival and the selected process continjues until it compltes
- Round Robin: Each process receives a fixed time quantum.ehen the time quantum expires,an unfinished process is moved back to the queue
- Priority Scheduling:The scheduler selects a process according to its priority. The priority rule will be defined consistently during implementation

4. Scheduler Components

The scheduler will interact with the following components:

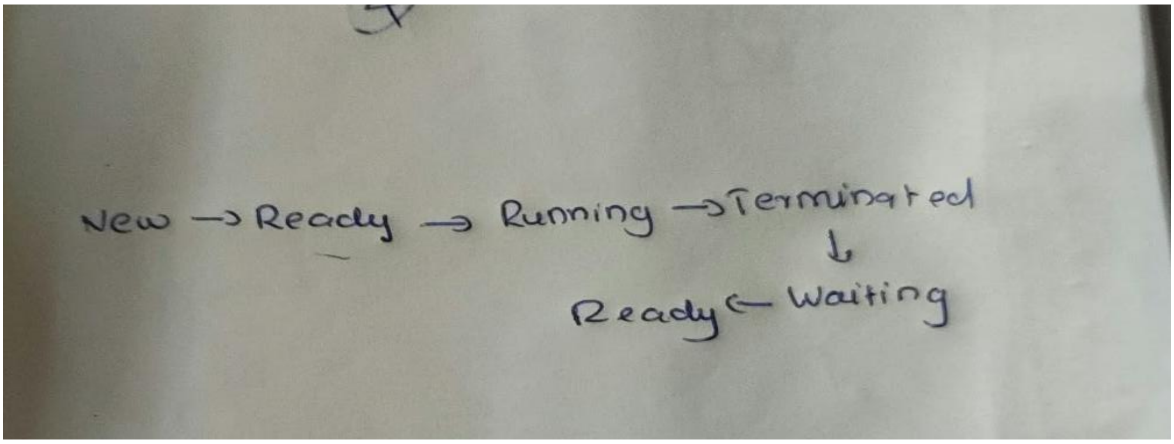
- Process Control Block (PCB)
- Ready Queue / Circular Queue● CPU / Processor state
- Program Counter and registers
- Scheduling algorithm

- Context-switching mechanism

5. Process States

Each process will have a state that represents its current condition in the simulator.
NEW – Process has been created.
READY – Process is waiting for CPU execution.
RUNNING – Process is currently executing on the CPU.
WAITING – Process is temporarily waiting for an event or operation.
TERMINATED – Process has completed execution.

6. Process State Flow



7. Ready Queue

The ready queue stores processes that are ready to use the CPU. The order of processes depends on the selected scheduling algorithm.

[P1] → [P2] → [P3] → [P4]

For Round Robin, an unfinished process can be placed at the end of the queue after its time quantum expires.

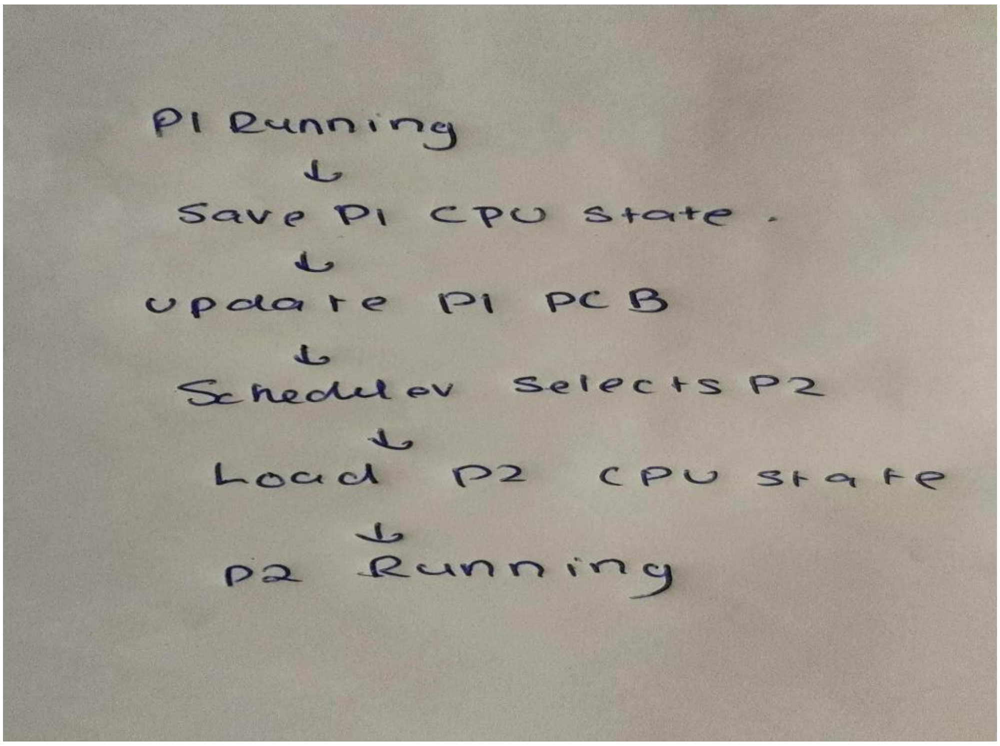
Content switching

it is the process of changing the CPU from one running process to another. The current process information is saved and the next selected process information is loaded.

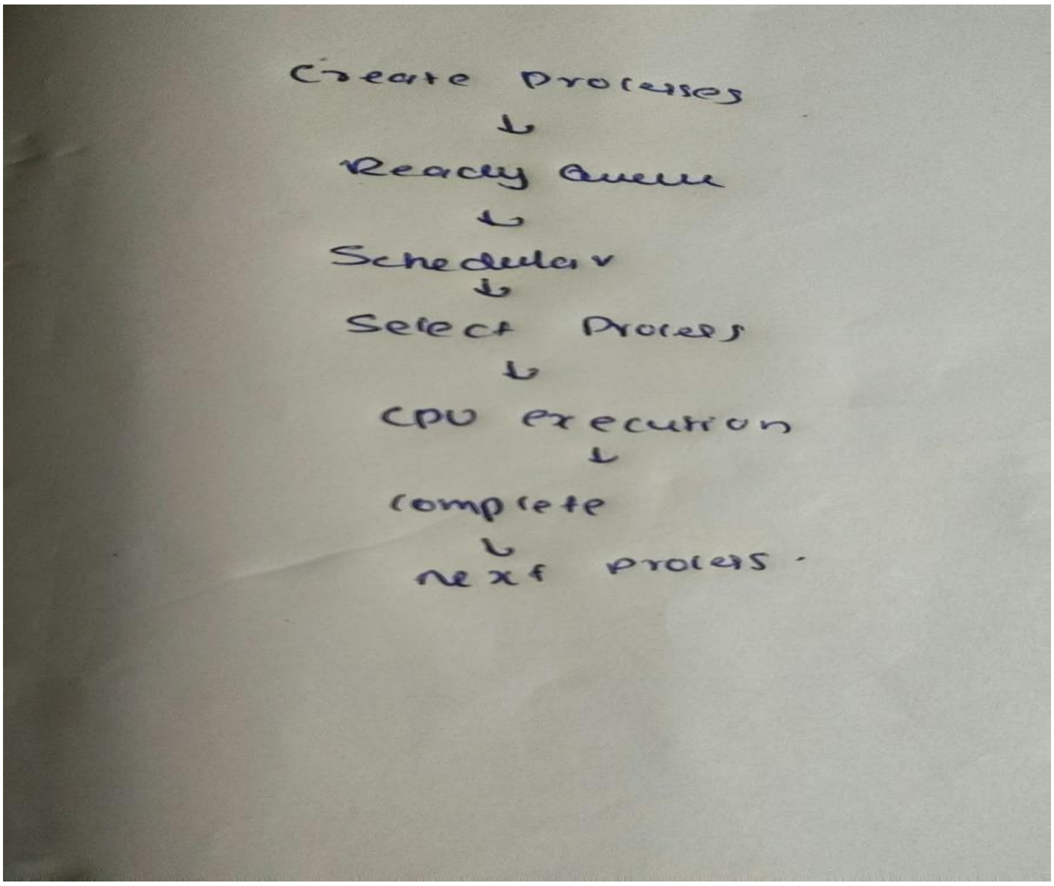
9. Context-Switching Steps

1. Stop or preempt the currently running process.
2. Save its CPU state and relevant information in the PCB.
3. Update the process state.
4. Select the next process from the ready queue.
5. Load the next process state.
6. Set the selected process to RUNNING.
7. Continue CPU execution.

10. Context-Switching Flow



11. Scheduling Flow



12. Scheduling Metrics

The simulator can later calculate Completion Time, Turnaround Time, Waiting Time, Response Time, Average Waiting Time, and Average Turnaround Time

13. Example Test Data

Process	Arrival	Burst	Priority
P1	0	5	2
P2	1	3	1
P3	2	4	3

FCFS: P1 → P2 → P3

Round Robin (example quantum = 2): P1 → P2 → P3 → P1 → P2 → P3 ...

14. UI Planning

The simulator UI should make the scheduler easy to understand by displaying:

- **List of processes**
- **Process state of each process**
- **Ready queue**
- **Selected scheduling algorithm**
- **Current running process**
- **Time quantum for Round Robin**
- **Context-switching information**
- **Scheduling results and metrics**

15. Implementation Plan

Step 1: Define Process and PCB data structures.

Step 2: Implement the ready queue.

Step 3: Implement FCFS scheduling.

Step 4: Implement Round Robin with time quantum.

Step 5: Implement Priority Scheduling.

Step 6: Implement context switching.

Step 7: Connect scheduler with CPU and process management.

Step 8: Add UI and test all scheduling algorithms.

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